



Statistics of Financial Markets

Seminar 16: Review



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- AI: Artificial Intelligence
- AUC: Area Under the (ROC) Curve
- BET: Bucharest Exchange Trading (index)
- EAD: Exposure At Default
- EODHD: EOD Historical Data (market-data provider)
- ES: Expected Shortfall
- EWMA: Exponentially Weighted Moving Average
- FHS: Filtered Historical Simulation
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- HS: Historical Simulation
- JB: Jarque-Bera (test)
- LGD: Loss Given Default
- MDD: Maximum DrawDown
- PD: Probability of Default
- ROC: Receiver Operating Characteristic (curve)
- SE: Standard Error
- SR: Sharpe Ratio
- VR: Variance Ratio

Today's Question and Route

- **Question:** can you solve, on paper and on data, a problem from any chapter of the course?
 - ▶ this seminar comes **before** Lecture 16: the section “Prerequisites for Today” is a formula sheet for all chapters
- **Route**
 - ▶ Part A: six exam-type problems on paper, from returns to systemic risk
 - ▶ Part B: three data tasks on the BET, the S&P 500 and Bitcoin, each with an interpretation question
 - ▶ Part C: an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Topic	Type	Chapters
A1	returns, drawdown, annualisation, Sharpe ratio	Solved	1
A2	Jarque–Bera, Student- t , Hill, stable scaling	Proposed	2, 3, 5
A3	Ljung–Box, variance ratio, robust test	Solved	7
A4	GARCH, EWMA and Parkinson volatility	Proposed	8, 9
A5	VaR 1%, ES 2.5%, Kupiec, traffic light	Proposed	10
A6	logit PD, expected loss, AUC, ΔCoVaR	Proposed	12, 15
B1	a full check-up of the BET since 2015	Solved	1–11
B2	VaR 1% backtests: S&P 500 and Bitcoin	Proposed	8, 10, 14
B3	efficiency and memory in two windows	Proposed	7, 11
C2	audit an AI answer	Proposed	0–15

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Price used	Period
BET, S&P 500	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$, each series on its own calendar; the last day is 18 September 2026
- Part A needs only a calculator; Part B uses the notebook, with the functions of the course already written (returns, moments, ljung_box, variance_ratio, garch_t, kupiec, hurst_rs)

Prerequisites for Today (1/6): Returns, Distributions, Tails

Quantity	Formula
Returns	$R_t = P_t/P_{t-1} - 1$, $r_t = \ln(1 + R_t)$; total: $e^{\sum r_t} - 1$
Annualisation	mean $\times A$, volatility $\times \sqrt{A}$; $A = 252$ (shares), 365 (crypto)
Sharpe, drawdown	$(\mu - r_f)/\sigma$, SE $\sqrt{(1 + \text{SR}^2/2)/Y}$; $D_t = P_t / \max_{s \leq t} P_s - 1$, recovery $d/(1 - d)$
Jarque–Bera	$\frac{n}{6} S^2 + \frac{n}{24} K^2 \sim \chi^2(2)$, $\chi_{0.95}^2(2) = 5.99$
Student- $t(\nu)$	excess kurtosis $6/(\nu - 4)$; unit-variance quantile $t_\nu^{-1}(\alpha) \sqrt{(\nu - 2)/\nu}$
Hill	$\hat{\alpha} = [\frac{1}{k} \sum_{i=1}^k \ln(L_{(i)}/L_{(k+1)})]^{-1}$, SE $\hat{\alpha}/\sqrt{k}$; moments exist for $p < \alpha$
Stable law	sum of h copies: scale $\times h^{1/\alpha}$ (Normal: $\sqrt{h}$)

Prerequisites for Today (2/6): Notation for Returns, Distributions and Tails

▣ Returns and performance

- ▶ P_t : the price at the end of day t ; R_t : the simple return; r_t : the log return; A : the number of observations per year
- ▶ μ , σ : the annual mean and volatility; r_f : the risk-free rate; SR: the Sharpe ratio; SE: its standard error; Y : the number of years
- ▶ D_t : the drawdown; $\max_{s \leq t} P_s$: the highest price up to day t ; d : the size of a loss, as a fraction

▣ Distributions and tails

- ▶ n : the sample size; S : skewness; K : excess kurtosis (both 0 for the Normal distribution); $\chi_{0.95}^2(2)$: the 5% critical value
- ▶ ν : degrees of freedom; $t_\nu^{-1}(\alpha)$: the α -quantile of the Student- t ; $\sqrt{(\nu - 2)/\nu}$ rescales it to variance 1
- ▶ $L_{(i)}$: the i -th largest loss; k : the number of tail losses; p : the order of a moment; h : the number of days summed

Prerequisites for Today (3/6): Efficiency, Volatility, Memory

Quantity	Formula
Ljung–Box	$Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi^2(m), \chi_{0.95}^2(4) = 9.49$
Variance ratio	$VR(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \hat{\rho}_k; Z = (VR - 1) / \sqrt{2(2q-1)(q-1)/(3qT)}$
Robust VR test	$Z^* = (VR - 1) / SE_{rob}$; reject at 5% if $ Z^* > 1.96$
EWMA	$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1 - \lambda) r_t^2, \lambda = 0.94$
GARCH(1,1)	$\sigma_{t+1}^2 = \omega + \alpha \varepsilon_t^2 + \beta \sigma_t^2, \bar{\sigma}^2 = \omega / (1 - \alpha - \beta), h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$
GARCH forecast	$E_t \sigma_{t+h}^2 = \bar{\sigma}^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \bar{\sigma}^2)$
Parkinson	$\sigma^2 = (\ln H - \ln L)^2 / (4 \ln 2)$
Hurst	$(R/S)_n \propto n^H$; risk over h days $\propto h^H$

Prerequisites for Today (4/6): Notation for Efficiency, Volatility and Memory

□ Dependence in time

- ▶ $\hat{\rho}_k$: the autocorrelation at lag k ; T : the number of returns; m : the number of lags tested; $\chi^2(m)$: the distribution of $Q(m)$ without autocorrelation
- ▶ q : the horizon of the variance ratio, in days; $VR = 1$ under a random walk; SE_{rob} : a standard error robust to volatility clustering

□ Volatility

- ▶ σ_t^2 : the variance forecast for day t ; λ : the EWMA decay factor; ε_t : the shock (the demeaned return)
- ▶ ω, α, β : GARCH parameters; $\bar{\sigma}^2$: the long-run variance; $h_{1/2}$: the half-life in days; E_t : the forecast made on day t for h days ahead
- ▶ H, L : the high and the low of the day (Parkinson)

□ Memory

- ▶ $(R/S)_n$: the rescaled range on blocks of n days; H : the Hurst exponent, 0.5 without memory

Prerequisites for Today (5/6): Risk, Backtesting, Credit, Systemic Risk

Quantity	Formula
VaR, ES	$\text{VaR}_\alpha = -q_\alpha > 0$; Normal: $-(\mu + \sigma z_\alpha)$, ES $-\mu + \sigma \varphi(z_\alpha)/\alpha$
Values	$z_{0.01} = -2.326$, $z_{0.025} = -1.960$, $\varphi(1.96) = 0.0584$, $t_5^{-1}(0.01) = -3.365$
Kupiec	$LR_{uc} = -2[(n-x)\ln(1-\alpha) + x\ln\alpha - (n-x)\ln(1-\hat{\pi}) - x\ln\hat{\pi}] \sim \chi^2(1)$, $\chi_{0.95}^2(1) = 3.84$
Traffic light (250 days)	green 0–4, yellow 5–9, red 10 or more exceptions
Logit, EL	$PD = 1/(1 + e^{-\eta})$, odds ratio e^{β_j} ; EL = PD $\times$ LGD $\times$ EAD
AUC, Gini	Gini = 2 AUC – 1
CoVaR	1% quantile regression $X_{sys} = a + bX_i$: CoVaR = $-(a + b q_\alpha^i)$, $\Delta\text{CoVaR} = b(q_{50}^i - q_\alpha^i)$

Prerequisites for Today (6/6): Notation for Risk, Credit and Systemic Risk

□ Risk and backtesting

- ▶ q_α : the α -quantile of the return; μ, σ : daily mean and standard deviation; z_α : the standard Normal quantile; φ : the standard Normal density
- ▶ $t_5^{-1}(0.01)$: the 1% quantile of a Student- t with 5 degrees of freedom
- ▶ x : the exceptions in n days; α : the target rate; $\hat{\pi} = x/n$: the observed rate; reject if $LR_{uc} > 3.84$

□ Credit

- ▶ $\eta = \beta_0 + \beta_1 x_1 + \dots$: the linear score; PD: probability of default; e^{β_j} : the factor by which a one-unit rise in x_j multiplies the odds
- ▶ EL: expected loss; LGD: loss given default; EAD: exposure at default; AUC: area under the ROC curve (0.5: no discrimination)

□ Systemic risk

- ▶ X_{sys}, X_i : the returns of the system and of bank i ; a, b : intercept and slope; q_α^i, q_{50}^i : the α -quantile and the median of the bank

A1: Returns, Drawdown and the Sharpe Ratio [Solved]

Task

- A share closes at 50.0, 51.5, 49.8, 52.3, 47.9 and 50.6 RON on six consecutive days.
- 1. Compute the five simple and the five log returns, in %.
- 2. Compute the total return from the prices and from $\sum_t r_t$.
- 3. Compute the maximum drawdown and the gain needed to return to the peak.
- 4. A fund has a daily mean return of 0.04% and a daily standard deviation of 1.3%; with $r_f = 3\%$ a year, compute its annual mean, its annual volatility, its Sharpe ratio and the SE of the Sharpe ratio over 10 years.
- Report: fourteen numbers and one sentence.

Solution

- 1. R : 3.00, -3.30, 5.02, -8.41, 5.64; r : 2.96, -3.36, 4.90, -8.79, 5.48
- 2. $50.6/50 - 1 = 1.20\%$; $\sum r_t = 1.19\%$ and $e^{1.19/100} - 1 = 1.20\%$
- 3. $MDD = 47.9/52.3 - 1 = -8.41\%$; gain $52.3/47.9 - 1 = 9.19\%$
- 4. $252 \times 0.04 = 10.08\%$; $\sqrt{252} \times 1.3 = 20.64\%$; $SR = (10.08 - 3)/20.64 = 0.343$; $SE = \sqrt{(1 + 0.343^2/2)/10} = 0.33$
- Interpretation: the Sharpe ratio is about one SE from zero: ten years of data cannot show that this fund beats the risk-free rate.

A2: Normality, Tails and Stable Scaling [Proposed]

Solution discussed in class

Task

- ▣ $n = 2000$ daily returns: skewness $S = -0.8$, excess kurtosis $K = 9$. The six largest losses, in %: 11.2, 8.4, 7.7, 6.0, 5.5, 4.9. Model: Seminar 2, A7 and Seminar 5, A1.
- ▣ 1. Compute JB, its skewness and kurtosis parts, and decide at 5%.
- ▣ 2. Compute ν of a Student- t distribution with the same excess kurtosis.
- ▣ 3. Compute the Hill estimate $\hat{\alpha}$ with $k = 5$ and its SE.
- ▣ 4. A stable law with $\alpha = 1.6$ describes daily returns: compute the 20-day scale factor and compare it with $\sqrt{20}$.
- ▣ Report: seven numbers, one decision and one sentence.

A3: Autocorrelation and the Variance Ratio [Solved]

Task

- ▣ $T = 1500$ daily returns; $\hat{\rho}_1 = -0.05$, $\hat{\rho}_2 = 0.03$, $\hat{\rho}_3 = 0.02$, $\hat{\rho}_4 = -0.01$; the robust SE of $\widehat{VR}(5)$ is 0.055.
- ▣ 1. Compute the Ljung–Box statistic $Q(4)$ and decide at 5%.
- ▣ 2. Compute $VR(5)$.
- ▣ 3. Compute $Z(5)$ and $Z^*(5)$ and decide at 5%.
- ▣ Report: four numbers, two decisions and one sentence.

Solution

- ▣ 1. $\sum \hat{\rho}_k^2 = 0.0039$; $Q(4) = 5.86 < 9.49$ ($p = 0.210$): not rejected
- ▣ 2.
 $1 + 2(-0.040 + 0.018 + 0.008 - 0.002) = 1 + 2 \times (-0.016) = 0.968$
- ▣ 3. $SE = \sqrt{72/22\,500} = 0.0566$; $Z = -0.57$;
 $Z^* = (0.968 - 1)/0.055 = -0.58$; both $|Z| < 1.96$: not rejected
- ▣ Interpretation: no evidence against the random walk at one week; weak-form efficiency is not rejected.

A4: Volatility Forecasts [Proposed]

Solution discussed in class

Task

- ▣ GARCH(1,1), returns in %: $\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$; today $\sigma_t^2 = 2.5$ and $\varepsilon_t = 1.0$. One day of an index: high 105.2, low 101.6. Model: Seminar 9, A1 and A3; Seminar 8, A3 and A5.
- ▣ 1. Compute σ_{t+1}^2 and the EWMA forecast with $\lambda = 0.94$.
- ▣ 2. Compute the persistence, the half-life, the long-run variance and the long-run annual volatility.
- ▣ 3. Compute $E_t[\sigma_{t+5}^2]$.
- ▣ 4. Compute the Parkinson variance of the day and the annualised Parkinson volatility.
- ▣ Report: eight numbers and one sentence.

A5: VaR, ES and Backtesting [Proposed]

Solution discussed in class

Task

- ▣ A position of 2 000 000 RON; daily returns with mean 0.05% and standard deviation 1.2%. Model: Seminar 10, A1, A3 and A5.
- ▣ 1. Compute the Normal VaR 1% and ES 2.5%, in % and in RON.
- ▣ 2. Compute VaR 1% if the standardised returns are Student- t with $\nu = 5$.
- ▣ 3. A model has 17 exceptions of VaR 1% in 1000 days: compute LR_{UC} and decide at 5%.
- ▣ 4. Give the Basel zone of 11 exceptions in 250 days.
- ▣ Report: seven numbers, two decisions and one sentence.

A6: Credit Scoring and CoVaR [Proposed]

Solution discussed in class

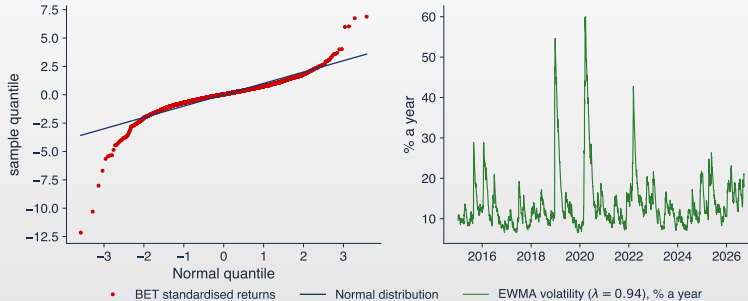
Task

- Logit: $\eta = -1.8 + 0.9x_1 - 0.6x_2$, applicant with $x_1 = 1.2$, $x_2 = 2.5$; LGD = 40%, EAD = 150 000 RON. A 1% quantile regression of the system return on a bank: $a = -1.8$, $b = 0.45$; bank: $q_{0.01} = -6.0\%$, median 0.05%. Model: Seminar 12, A1 and A5; Seminar 15, A1.
- 1. Compute the PD of the applicant and the odds ratio of x_1 .
- 2. Compute the expected loss of the loan.
- 3. A scorecard has Gini = 0.56: compute its AUC.
- 4. Compute CoVaR 1% at the bank's VaR and at its median, and ΔCoVaR .
- Report: seven numbers and one sentence.

B1: A Full Check-Up of the BET [Solved]

- ▣ **Question:** what do [Chapters 1–11](#) say about the BET since 2015?
- ▣ **Inputs:** BET closes since 2 January 2015, daily log returns in %
- ▣ **Tasks**
 1. Compute the annual mean, the volatility, the Sharpe ratio with its SE and the maximum drawdown.
 2. Compute the skewness, the excess kurtosis, JB and the Hill tail index of the losses with $k = 2.5\%$ of n .
 3. Compute Ljung–Box $Q(10)$ on r_t and on r_t^2 , and fit a GARCH(1,1)- t .
 4. Compute VaR 1% and ES 2.5% by historical simulation and with the Normal distribution.
 5. Interpretation: which three facts would you write in an exam answer about the BET?
- ▣ **Report:** a table of about fifteen numbers, the chart and three sentences
- ▣ **Notebook:** section B1

B1: Solution (1/2) [Solved]



- $n = 2925$, 6 January 2015 – 18 September 2026; critical value $\chi^2_{0.95}(10) = 18.31$; EWMA volatility peak 60.0% on 25 March 2020



B1: Solution (2/2) [Solved]

Performance		Distribution, dependence	
mean, % a year	13.2	skewness; excess kurtosis	−1.41; 17.8
volatility, % a year	15.5	JB; Hill $\hat{\alpha}$ (SE)	39 794; 2.19 (0.26)
Sharpe (SE)	0.93 (0.35)	$Q(10): r_t; r_t^2$	54.7; 701
MDD	−31.1%	GARCH- $t: \alpha + \beta; h_{1/2}$	0.925; 8.8
VaR 1%: HS; Normal	2.72; 2.23	ES 2.5%: HS; Normal	3.21; 2.24

- Interpretation: (1) heavy tails: the Normal VaR 1% is 18% too low; (2) returns are weakly autocorrelated, squared returns strongly: volatility clustering; (3) the Sharpe ratio is less than three SE from zero

B2: Backtesting VaR 1% for the S&P 500 and Bitcoin [Proposed]

- **Question:** does a heavy-tailed method beat a Normal method with fast volatility? Model: B1 and Seminar 10, B3.
- **Inputs:** S&P 500 and Bitcoin, all data; forecasts for every day since 1 January 2017
- **Tasks**
 1. Compute the one-day VaR 1% by historical simulation on the last 500 returns.
 2. Compute the one-day EWMA-Normal VaR 1% with $\lambda = 0.94$.
 3. Count the exceptions of each method and run the Kupiec test.
 4. Find the largest number of exceptions in any 250 consecutive days.
 5. Interpretation: which method would a supervisor accept for each series?
- **Report:** one table and two sentences
- **Notebook:** section B2

B3: Efficiency and Memory in Two Windows [Proposed]

- **Question:** did the BET and the S&P 500 become more efficient after 2015? Model: A3; Seminar 7, B1 and Seminar 11, B1.
- **Inputs:** BET and S&P 500, two windows: 2000–2014 and 2015–2026
- **Tasks**
 1. Compute $VR(5)$ and the robust $Z^*(5)$ in each window.
 2. Compute the Hurst exponent (R/S) of r_t and of $|r_t|$ in each window.
 3. Interpretation: did the BET become more efficient after 2015?
- **Report:** one table of sixteen numbers and two sentences
- **Notebook:** section B3

C2: Audit an AI Answer [Proposed]

- ▣ A student asked an AI assistant to summarise the course for the exam, with the course data. The answer:
- ▣ (a) The VaR 99% of the BET since 2015 is -2.72%.
- ▣ (b) The BET returns are close to Normal: the skewness is only -1.41.
- ▣ (c) Bitcoin has a daily standard deviation of 3.49%, so its annual volatility is 55.4%.
- ▣ (d) For the S&P 500, $\alpha + \beta = 0.993$, so a volatility shock halves in $1/(1 - 0.993) = 139$ days.
- ▣ (e) The Hurst exponent of $|r|$ is 0.84 for the S&P 500, so its returns are predictable and the market is inefficient.
- ▣ (f) For Normal returns, ES 2.5% and VaR 1% are almost the same number.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- ▣ Every exam problem has three parts: the formula with the numbers, the result with its unit and sign, the interpretation
- ▣ Tests: state H_0 , the statistic, the critical value, the decision; prefer the robust version
- ▣ Risk: VaR 1% is a positive loss; the Normal distribution understates it on real data
- ▣ Volatility is predictable, the sign of returns almost not
- ▣ An AI answer is a draft: check the level and the sign of VaR, the annualisation factor and every formula

After the Seminar

- Lecture 16 gathers the course: the course map, two recap slides per chapter, the toolbox, eight solved exam-type problems, the project
- Try the [Proposed] tasks on paper first, then check them in the notebook
- B3 can grow into a team project: rolling windows, Monte Carlo bands, more emerging markets
- Reading: [Franke, Härdle & Hafner \(2019\)](#); exercises in [Borak, Härdle & López-Cabrera \(2013\)](#)
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

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